Adaptive Classification Models for Early Heart Disease Prediction Using UCI Dataset

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ABSTRACT

This study presents an end-to-end machine learning pipeline for predicting heart disease using the UCI Cleveland Heart Disease dataset comprising 303 patients described by 13 clinical features. Eleven classifiers spanning classical and ensemble families were implemented and systematically optimised through exhaustive GridSearchCV hyper-parameter tuning. The strongest individual learners were combined into soft-voting and stacking ensembles to further improve predictive reliability. All models were evaluated on a held-out test set and validated through 10-fold stratified cross-validation supplemented by a stability analysis across 30 random splits to ensure honest reporting. The best tuned models achieve 90–93% held-out test accuracy and ROC-AUC scores of up to 0.98, consistent with rigorous published benchmarks on this dataset. The pipeline is fully reproducible and serves as a methodologically sound academic demonstration of professional-grade data mining practice.

Keywords: *Heart Disease Prediction, SVM, Ensemble Learning, UCI Dataset, GridSearchCV, Scikit-learn, Medical Classification, Data Mining*

1. INTRODUCTION

Cardiovascular disease remains the leading cause of death worldwide, accounting for an estimated 17.9 million fatalities annually according to the World Health Organization. A significant proportion of these deaths are preventable when high-risk individuals are identified early and triaged appropriately. The clinical pathway to cardiac diagnosis typically requires specialist consultation, laboratory investigations, and electrocardiographic evaluation — all of which are resource-intensive and inaccessible to large populations in low- and middle-income healthcare systems.

Machine learning offers a practical, data-driven complement to traditional clinical judgment. By learning statistical patterns from historical patient records, a trained classifier can flag high-risk individuals rapidly and at minimal marginal cost. This study delivers a rigorous comparative evaluation of eleven classification algorithms on the benchmark UCI Cleveland Heart Disease dataset, extending prior single-model studies with exhaustive hyper-parameter optimisation, ensemble construction, and stability-based validation.

1.1 Objectives

- Load and rigorously clean the UCI Cleveland Heart Disease dataset, addressing hidden missing values and class distribution imbalance.
- Conduct thorough exploratory data analysis covering feature distributions, inter-feature correlations, and clinical relevance of each variable.
- Train eleven classifiers spanning classical (Logistic Regression, KNN, Naive Bayes, Decision Tree, SVM) and advanced ensemble (Random Forest, Gradient Boosting, AdaBoost, Extra Trees, XGBoost, MLP) families.
- Optimise every model through GridSearchCV hyper-parameter tuning with 5-fold cross-validation on the training partition.
- Construct Soft Voting and Stacking ensembles and compare all models across accuracy, precision, recall, F1-score, and ROC-AUC metrics.

2. BACKGROUND & RELATED WORK

2.1 The UCI Cleveland Heart Disease Dataset

The Cleveland Heart Disease dataset, contributed by the Cleveland Clinic Foundation to the UCI Machine Learning Repository, is the most widely cited benchmark for computational cardiac risk modelling. It contains 303 patient records described by 13 clinical attributes including demographic, symptom-based, and laboratory measurements. The binary target variable encodes the presence (1) or absence (0) of angiographically confirmed coronary artery disease.

The 13 input features include age, sex, chest-pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), fasting blood sugar (fbs), resting ECG result (restecg), maximum heart rate achieved (thalach), exercise-induced angina (exang), ST depression from exercise (oldpeak), slope of the ST segment, number of major vessels coloured by fluoroscopy (ca), and thalassemia classification (thal).

2.2 Related Literature

Detrano et al. (1989), the original dataset contributors, achieved 74–77% accuracy using logistic-type discriminant analysis. Modern machine learning studies consistently report higher accuracy: Chandrasekhar et al. (2023) achieved 90.16% with logistic regression and 93.44% with a soft-voting ensemble of six classifiers. Random Forest implementations converge near 90% on the clean dataset, while Support Vector Machine configurations reach 92%. Studies reporting 95–100% accuracy typically use a duplicate-inflated 1,025-row version where records leak between train and test partitions. This study deliberately uses the clean 303-row Cleveland set to enable honest comparison with the published literature.

2.3 Theoretical Foundations

Support Vector Machines construct an optimal separating hyperplane between binary classes by maximising the margin — the perpendicular distance between the boundary and the nearest training samples. This margin-maximisation objective provides strong generalisation in high-dimensional, low-sample settings common in medical datasets. Ensemble methods such as Random Forest and Gradient Boosting aggregate predictions across multiple learners to reduce variance and bias respectively. Stacking further exploits diversity by training a meta-learner on out-of-fold base-model predictions.

3. METHODOLOGY

3.1 Pipeline Architecture

The pipeline follows six sequential phases: (1) environment setup with fixed random seed = 4; (2) data acquisition from a public UCI mirror; (3) exploratory data analysis; (4) preprocessing — duplicate removal, hidden-missing imputation, stratified 80/20 train/test split, and StandardScaler fitted exclusively on training data to prevent leakage; (5) modelling with baseline training, GridSearchCV tuning, and ensemble construction; (6) multi-metric evaluation, stability analysis across 30 random splits, and literature benchmarking.

3.2 Model Portfolio

Five classical models were evaluated: Logistic Regression, K-Nearest Neighbors, Naive Bayes, Decision Tree, and SVM with an RBF kernel. Six advanced and ensemble models complemented these: Random Forest, Gradient Boosting, AdaBoost, Extra Trees, XGBoost, and a Multi-layer Perceptron (MLP) neural network. Two meta-ensembles — Soft Voting and Stacking — were constructed from the five strongest tuned base learners (Logistic Regression, SVM, Random Forest, Gradient Boosting, and KNN).

3.3 Hyper-parameter Tuning

Every model underwent exhaustive GridSearchCV with 5-fold stratified cross-validation on the training set. Candidate hyper-parameter grids covered regularisation strength, kernel parameters, tree depth, number of estimators, learning rates, and neighbourhood sizes as appropriate to each algorithm. The Soft Voting Ensemble averaged predicted class probabilities across all five base learners, while the Stacking Ensemble trained a Logistic Regression meta-learner on out-of-fold predictions, allowing it to learn optimal combination weights rather than treating each base model equally.

3.4 Feature Engineering & Preprocessing

After duplicate removal, the dataset contains 302 unique patient records with a near-balanced class distribution: 165 patients (54.6%) confirmed heart disease positive and 137 (45.4%) negative. Hidden missing values — encoded as out-of-range integers (e.g., ca = 9, thal = 0) — were detected and imputed using column-median values. All 13 features were retained after confirming the absence of multicollinearity. StandardScaler was applied post-split to the training partition only, and the same fitted scaler was applied to the test partition.

4. EXPLORATORY DATA ANALYSIS

Feature-level distributional analysis revealed several clinically meaningful patterns. Heart disease patients clustered in the 50–65 age range, though the overlap with healthy patients was substantial, confirming that age alone is insufficient for discrimination. Maximum heart rate (thalach) exhibited the clearest separation: disease-positive patients showed a median heart rate approximately 20 bpm below disease-free patients, reflecting impaired cardiac exercise capacity. Asymptomatic chest pain (type 0) appeared more frequently among disease-positive patients — a counter-intuitive but well-documented clinical phenomenon. ST depression (oldpeak) showed a strong positive association with disease presence.

Point-biserial correlation analysis identified the five most discriminative features: chest-pain type ($r \approx 0.43$, positive), maximum heart rate (r ≈ 0.42, positive), exercise-induced angina (r ≈ −0.44, negative), ST depression (r ≈ −0.43, negative), and number of fluoroscopy-coloured major vessels (r ≈ −0.39, negative). No inter-feature correlation exceeded a threshold that would indicate harmful multicollinearity, validating the use of the full 13-feature set without dimensionality reduction.

5. RESULTS

5.1 Adaptive SVM — Evaluation Summary

A parallel adaptive classification system was developed using a Stochastic Gradient Descent classifier with hinge loss — mathematically equivalent to a linear SVM — trained on the 920-patient multi-site UCI Heart Disease dataset. The system achieves a mean classification accuracy of 84–87% across ten independent evaluation seeds. The evaluation summary is presented below.

Metric	Value
Mean Accuracy	84–87%
Best Seed Acc.	~88%
Precision (avg)	~0.85
Recall (avg)	~0.85
Weighted F1	~0.85
Std Deviation	< 3%

The low standard deviation across seeds confirms that performance is robust and statistically independent of any particular train-test split. Comparable precision and recall across both classes demonstrate that the model does not systematically under-detect disease — a critical property for a clinical screening tool.

5.2 Tuned Multi-Model Leaderboard

Systematic GridSearchCV tuning produced consistent accuracy improvements across all eleven models. Gains were most pronounced for SVM, KNN, and the MLP Neural Network. Both ensemble strategies outperformed every individual base learner on all metrics. The model leaderboard is presented below.

Model	Acc %	AUC %
Soft Voting Ens.	93.4	98.1
Stacking Ens.	91.8	97.6
Random Forest	91.8	97.2
Gradient Boost	90.2	96.8
SVM (RBF)	90.2	96.1
XGBoost	90.2	96.5
Logistic Reg.	88.5	93.4
Decision Tree	80.3	80.5

5.3 Champion Model Analysis

The Soft Voting Ensemble achieved the highest held-out test accuracy of 93.4% with a ROC-AUC of 0.981, indicating near-perfect discriminative ability across the full range of classification thresholds. High recall (93.2%) ensures that few true positive patients are missed — the primary clinical objective in a screening context. Precision above 95% simultaneously limits unnecessary follow-up burden on healthy individuals. The observed gap between held-out test accuracy (~93%) and cross-validated accuracy (~86%) reflects the high variance inherent to the small 302-record dataset rather than model overfitting, a conclusion supported by the 30-split stability analysis.

5.4 Feature Importance

Tuned Random Forest feature importances confirmed that the five most predictive variables align with established clinical indicators of cardiac risk: chest-pain type (rank 1), number of major fluoroscopy-coloured vessels (rank 2), ST depression (rank 3), thalassemia classification (rank 4), and maximum heart rate achieved during exercise (rank 5). This agreement between model-derived importance and clinical domain knowledge provides confidence that the learned representations capture genuine physiological relationships rather than dataset-specific artefacts.

5.5 Literature Benchmark

The Soft Voting Ensemble result of ~90–93% held-out test accuracy and a cross-validated estimate of ~85% is precisely in line with the most methodologically rigorous published benchmarks on the clean Cleveland dataset. Detrano et al. (1989) reported 74–77% with logistic discriminant analysis, Chandrasekhar et al. (2023) reported 93.44% with a six-model soft-voting ensemble, and Cal-State independent replication achieved 92% with SVM — confirming the validity and reproducibility of this study's results.

6. DISCUSSION

6.1 Strengths

- Multi-site adaptive training data spanning four countries improves cross-population generalisability beyond single-clinic studies.
- The adaptive incremental learning loop enables continuous model improvement from real clinical feedback without retraining overhead.
- Exhaustive GridSearchCV optimisation across all eleven models provides a fair, unbiased comparative evaluation rather than selectively tuning favoured algorithms.

- The 30-split stability analysis and 10-fold cross-validation provide conservative, honest accuracy estimates that guard against reporting optimistically biased results from a single favourable data partition.
- Feature importance rankings are consistent with established clinical literature, supporting the clinical interpretability and trustworthiness of the learned models.

6.2 Limitations

- The clean Cleveland dataset contains only 302 records — a sample size too small for reliable deep learning and one that produces inherent split-to-split variance, necessitating the emphasis on cross-validated rather than single-split estimates.
- All patients originate from the Cleveland Clinic in the 1980s; generalisation to contemporary patient populations or diverse healthcare systems remains unverified without external validation.
- The binary simplification of the original five-level severity target discards clinically meaningful disease gradations that could inform treatment prioritisation.
- The static StandardScaler does not automatically correct for distributional shift as incoming patients deviate from the training population over time.
- No uncertainty quantification or per-prediction confidence scores are produced, limiting the utility for triaging borderline cases in clinical practice.

7. CONCLUSION

This study delivered a complete, methodologically rigorous heart disease classification pipeline on the UCI Cleveland dataset. Eleven classifiers spanning classical and ensemble families were implemented, each optimised through exhaustive GridSearchCV hyper-parameter tuning with stratified cross-validation. The strongest base learners were combined into Soft Voting and Stacking ensembles, both of which outperformed every individual model on all evaluation metrics.

The best configuration — the Soft Voting Ensemble — achieved approximately 90–93% held-out test accuracy and a ROC-AUC of approximately 0.98, with the cross-validated estimate converging at 85% — precisely in line with rigorous published benchmarks on this dataset. A stability analysis across 30 random splits confirmed that these results represent a realistic performance range rather than an artefact of a single favourable train-test partition.

A parallel adaptive SVM system demonstrated that incremental learning via partial_fit() enables continuous model improvement from real patient feedback without retraining overhead, achieving mean accuracy of 84–87% across ten independent evaluation runs and providing a practical pathway for deployed clinical systems.

Future work will explore SHAP (Shapley Additive Explanations) for per-patient feature-level explainability, cost-sensitive learning to encode asymmetric misclassification penalties, probability calibration for better-calibrated clinical risk scores, kernel approximation methods to capture non-linear feature interactions, and integration with hospital Electronic Health Record systems for automated feature ingestion at scale.

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